

FCC 3101 CHAPTER 4

THE ENGINEER AS A PROFESSIONAL

INTRODUCTORY

REMARKS

1 This chapter is divided into seven sections: -

1. Glossary
2. Engineering Profession
3. Professional Ethics
4. Design
5. Construction
6. Maintenance
7. Safe Demolition

2 This chapter shall be covered in **THREE** parts: -

Part 1 covers

Glossary

Engineering Profession

Professional Ethics

Part 2 covers

Design

Part 3 covers

Construction

Maintenance

Safe Demolition

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CHAPTER 4

PART 1

4.1 GLOSSARY

- 4.1.1 **Profession** means an occupation which requires a high level of **knowledge** and **skill**, and in which the **professionals** are bound by **professional ethics**.
- 4.1.2 **Professional ethics** are the **set standards** of **individual** (or corporate) **behavior** expected of a **professional** (or corporation) involved in professional work.
- 4.1.3 A **Professional** is a **person** who is qualified and licensed to practice and earn a living in a given discipline within a profession.

4.1.4 Professionalism refers to three aspects:

- (1) strict adherence to **professional ethics**.
- (2) high level of **skill** (speed and precision).
- (3) positive **attitude** (correct mindset).

4.1.5 Discipline means a specific **branch of knowledge** within a given profession. **Example: Civil, Electrical and Mechanical are disciplines within engineering.**

4.1.6 Engineer (Eng) is the title given to a person who is **qualified, registered and licensed** to practice and earn a living in one of the engineering disciplines.

4.1.7 **Engineer** may also refer to: -

- 1) A person who is pursuing a BSc degree in engineering in a recognized university (**student engineer**).
- 2) A person who has recently graduated in engineering from a recognized university (**graduate engineer**).

4.1.8 **Consultant** means a person who is consulted because of his or her high level of **knowledge, skill** and **experience** in a **specific discipline**.

4.1.9 **Consultant engineer** means an engineer whose occupation is consultancy work.

[The “opposite” of consultant engineer is **contractor engineer**].

4.2 ENGINEERING PROFESSION

4.2.1 Levels of Cadre

The engineering profession has four levels of cadre: -

1. **Artisans:** Masons, carpenters, welders, plumbers and electricians who hold government **Trade Test Certificates** from TVET colleges.
2. **Technicians:** Holders of **Ordinary National Diploma** from polytechnics, in an area of specialization within a given discipline.
3. **Technologists:** Holders of **Higher National Diploma** from polytechnics, in an area of specialization within a given profession.
4. **Engineers:** Holders (or prospective holders) of **BSc degree** from a recognized university in an engineering discipline.

4.2.2 Classification of Engineers

Engineers are classified into four categories: -

1. **Student engineer:** A person pursuing a BSc degree in engineering from a recognized university.
2. **Graduate engineer:** A person who has recently graduated with a BSc degree from a recognized university and is registered and licensed to work under the supervision of a professional engineer.
3. **Professional engineer:** A person who is registered and licensed to work and earn a living as a practicing engineer.
4. **Consultant engineer:** A person who is registered and licensed to work and earn a living as a consultant engineer.

4.2.3 Engineering Regulatory Bodies

The engineering profession is regulated by various bodies: -

1. **ESA (and ACES):** These are engineering student's associations at UoN.
2. **Engineers Board of Kenya (EBK)** is a statutory body which registers engineers.
3. **Institution of Engineers of Kenya (IEK)** is a members club for engineers.
4. **Association of Consultant Engineers (ACE)** is a members club for consultant engineers.

4.2.4 Construction Regulatory Bodies

The construction industry is regulated by various bodies: -

1. **Central Government** regulates the construction industry through its line ministries.
2. **County Governments** regulate projects which are within their jurisdiction.
3. **National Construction Authority (NCA)** regulates the construction process
4. **National Environmental Management Authority (NEMA)** regulates all projects.

4.3 PROFESSIONAL ETHICS

4.3.1 Introduction

- 4.3.1.1 Professional ethics refer to the standards of **personal behavior** (or corporate behavior) expected of a professional (or corporation).
- 4.3.1.2 Professionals use **specialist skills** to provide **service to the public**.
- 4.3.1.3 Professionals are required to provide service to the public in a manner which is **morally acceptable**.
- 4.3.1.4 Professional ethics is therefore a **moral issue**.

4.3.2 Origins

- 4.3.2.1 Professional ethics originated in the **divinity** and **military** professions around 1650.
- 4.3.2.2 The **clergy** were subjected to **vows of chastity** in order to avoid sexual activity (**immoral**).
- 4.3.2.3 **Recruits** to the army were subjected to **vows of loyalty** in order avoid disobedience (**immoral**).
- 4.3.2.4 Later, professional ethics was adopted in **law** and **medicine**.
- 4.3.2.5 By the time **civil engineering** was born in 1750, professional ethics was 100 years old.

4.3.3 Ethical Behavior

Ethical behavior is defined in terms of **ten** behavioral qualities: -

- 4.3.3.1 Honesty:** The quality of being **truthful, frank, in good faith** and **without malice**.
- 4.3.3.2 Objectivity:** The quality of being **unemotional, unbiased, unprejudiced**.
- 4.3.3.3 Accountability:** The quality of being **responsible** and **answerable** for your actions.
- 4.3.3.4 Transparency:** The quality of being **accessible to independent scrutiny**.

- 4.3.3.5 **Confidentiality:** The quality of being able to **keep client information** in secret.
- 4.3.3.6 **Respect:** The quality of being able to **hold other people** in **high regard**.
- 4.3.3.7 **Loyalty:** The quality of being **faithful** and **devoted** to the profession.
- 4.3.3.8 **Lawfulness:** The quality of being able to **obey the laws** of the land.
- 4.3.3.9 **Fair play:** The quality of being able to **follow the rules** in all your actions.
- 4.3.3.10 **Integrity:** Steadfast adherence to the code of conduct.

4.3.4 Enforcement

- 4.3.4.1 Professional ethics are enforced by strict adherence to a **code of conduct** for members of the profession.
- 4.3.4.2 Enforcement of professional ethics prevents **corrupt practices** and **exploitation of the client**. Enforcement is also used to protect the **reputation of the profession**.
- 4.3.4.3 Professionals who, for whatever reason, do not adhere to the code of conduct are subjected to a **disciplinary process** by the registration board.

4.3.5 Punishment

- 4.3.5.1 If a professional is found to have acted unprofessionally, he or she is subjected to heavy punishment which may include: -
1. Financial fines
 2. Suspension of practicing license
 3. Deregistration and withdrawal of license
 4. Civil or criminal prosecution

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END OF PART 1